

THE TRIGEMINAL NERVE

Dr. SARAVANAN

3 BDS LOCAL ANESTHESIA

SEMESTER 2 -Lecture 1

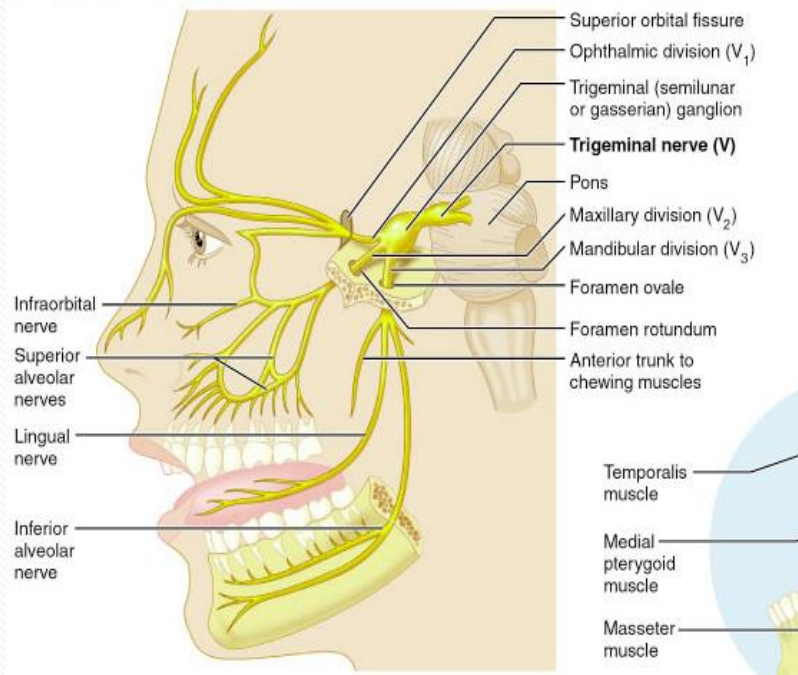
● The Trigeminal nerve

- The largest cranial nerve(V)
- It is mixed nerve (large sensory and small motor)
- The 3 divisions are

Ophthalmic V₁

Maxillary V₂

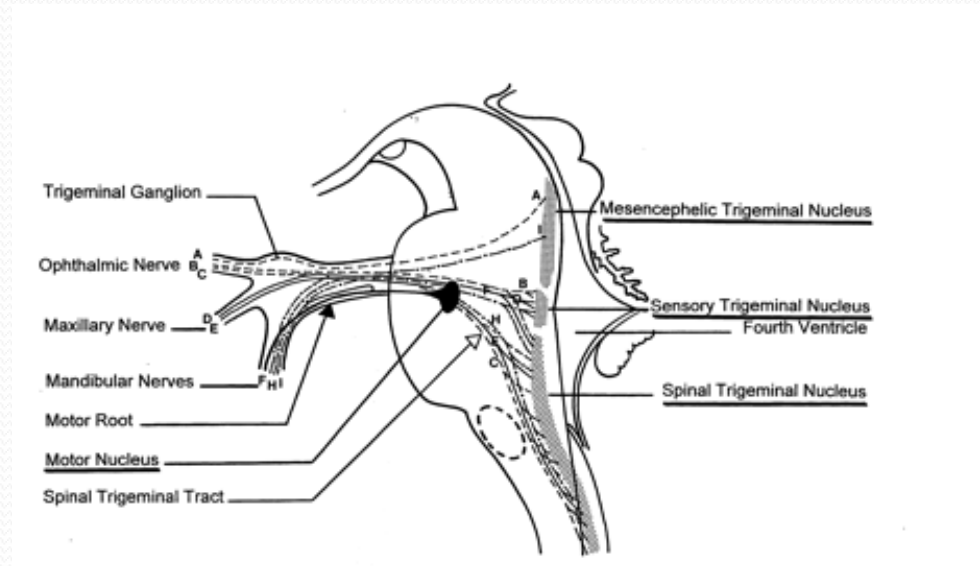
Mandibular V₃



MOTOR ROOT-COURSE AND DISTRIBUTION

- The motor root arises separately from the sensory root ,originating in the **motor nucleus within pons and medulla oblongata.**
- Its fibers forming a separate small nerve root travels anteriorly along with the main sensory root.

Just below the foramen ovale,the motor root unites with sensory root of mandibular division to form a single nerve trunk.

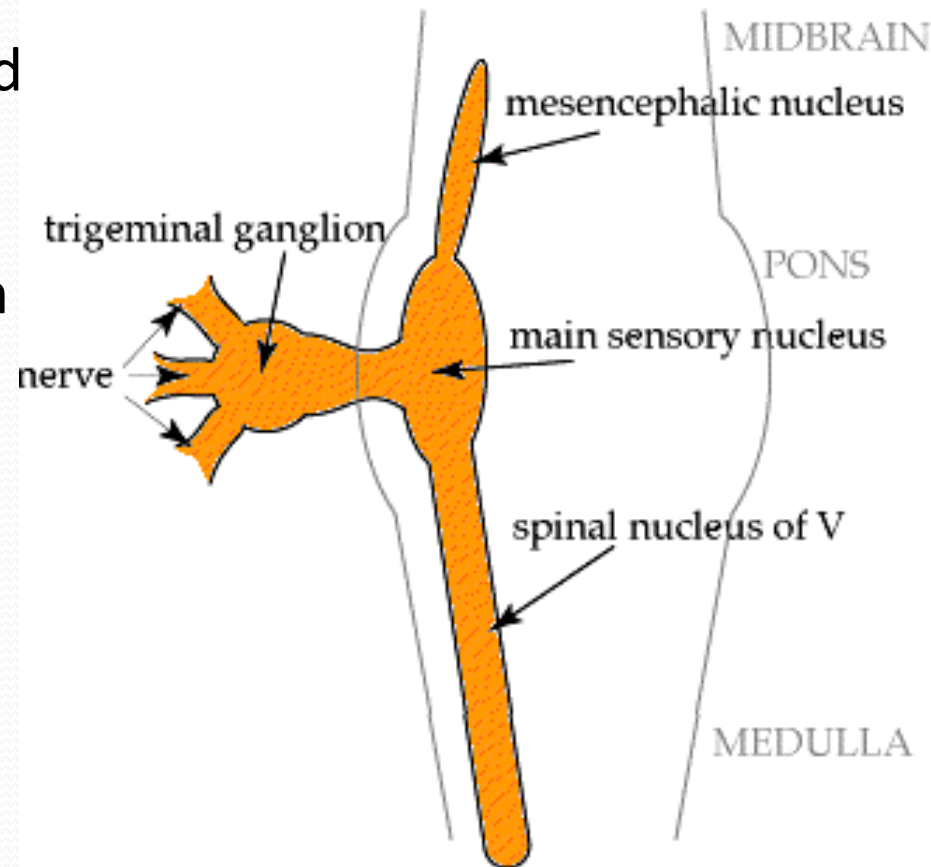


SENSORY NUCLEI

Mesencephalic nucleus - first order sensory nucleus situated in midbrain just lateral to aqueduct. pressure sensation

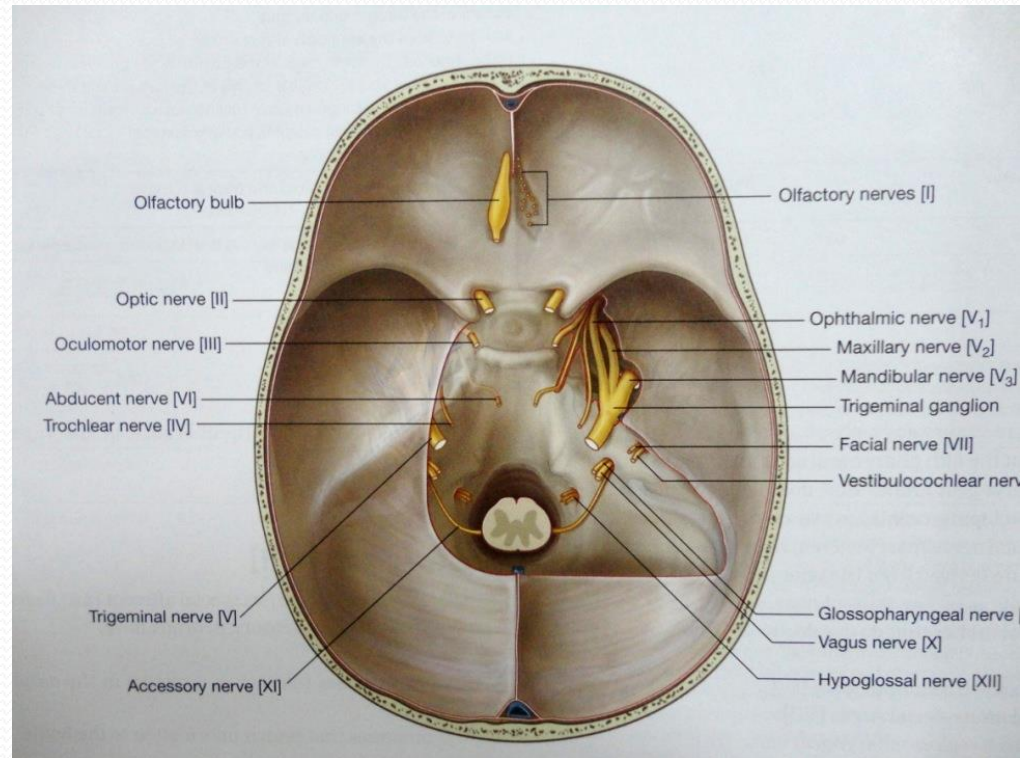
Principal sensory nucleus- lies in pons lateral to motor nucleus relays discriminative touch

Spinal nucleus- extends from caudal end of principal sensory nucleus .
Nucleus, pain and temperature relay

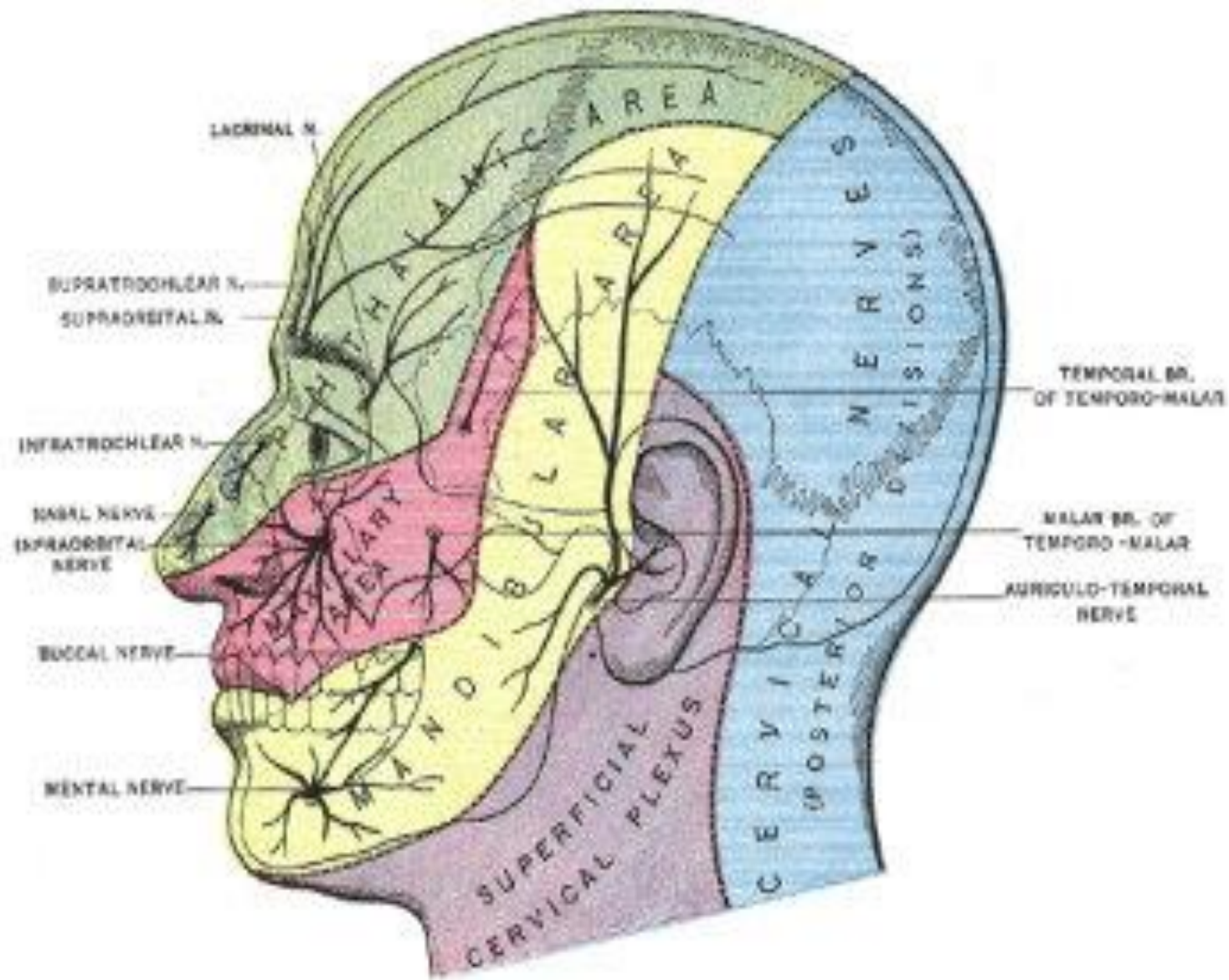


THE TRIGEMINAL GANGLION

- Also called as SEMILUNAR OR GASSERIAN GANGLION.
- Sensory ganglion.
- Crescentic in shape with convexity anterolaterally.
- **LOCATION:** lies in a bony fossa at apex of the petrous temporal bone on floor of middle cranial fossa , just lateral to posterior Part of lateral wall of the cavernous sinus.(meckel's cavity) 5 cm deep to the preauricular point.



Sensory innervations of Trigeminal nerve

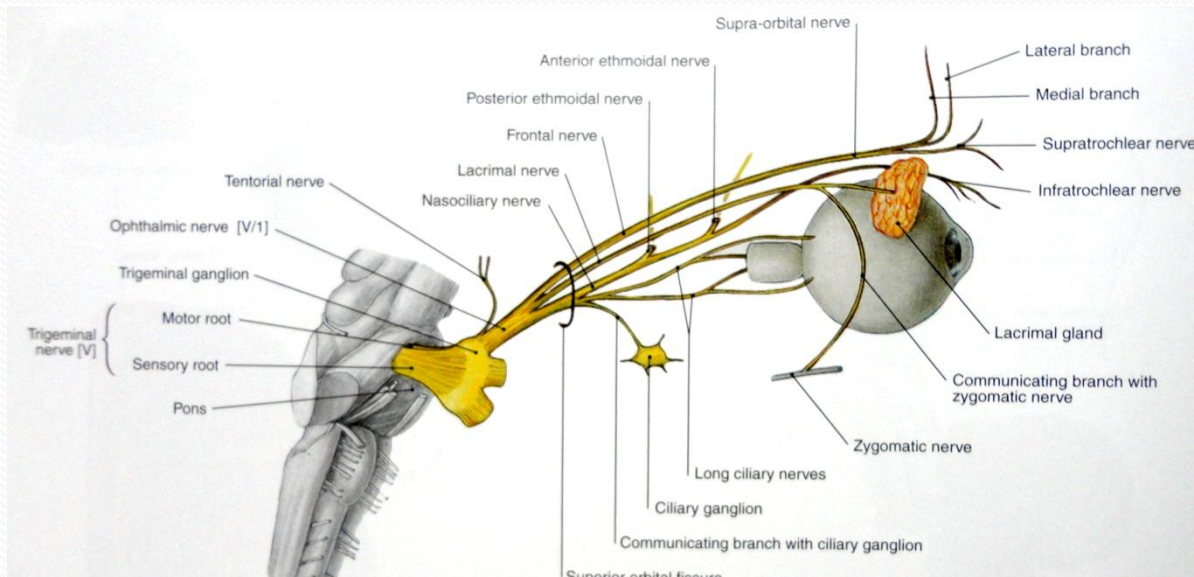


OPHTHALMIC NERVE V1

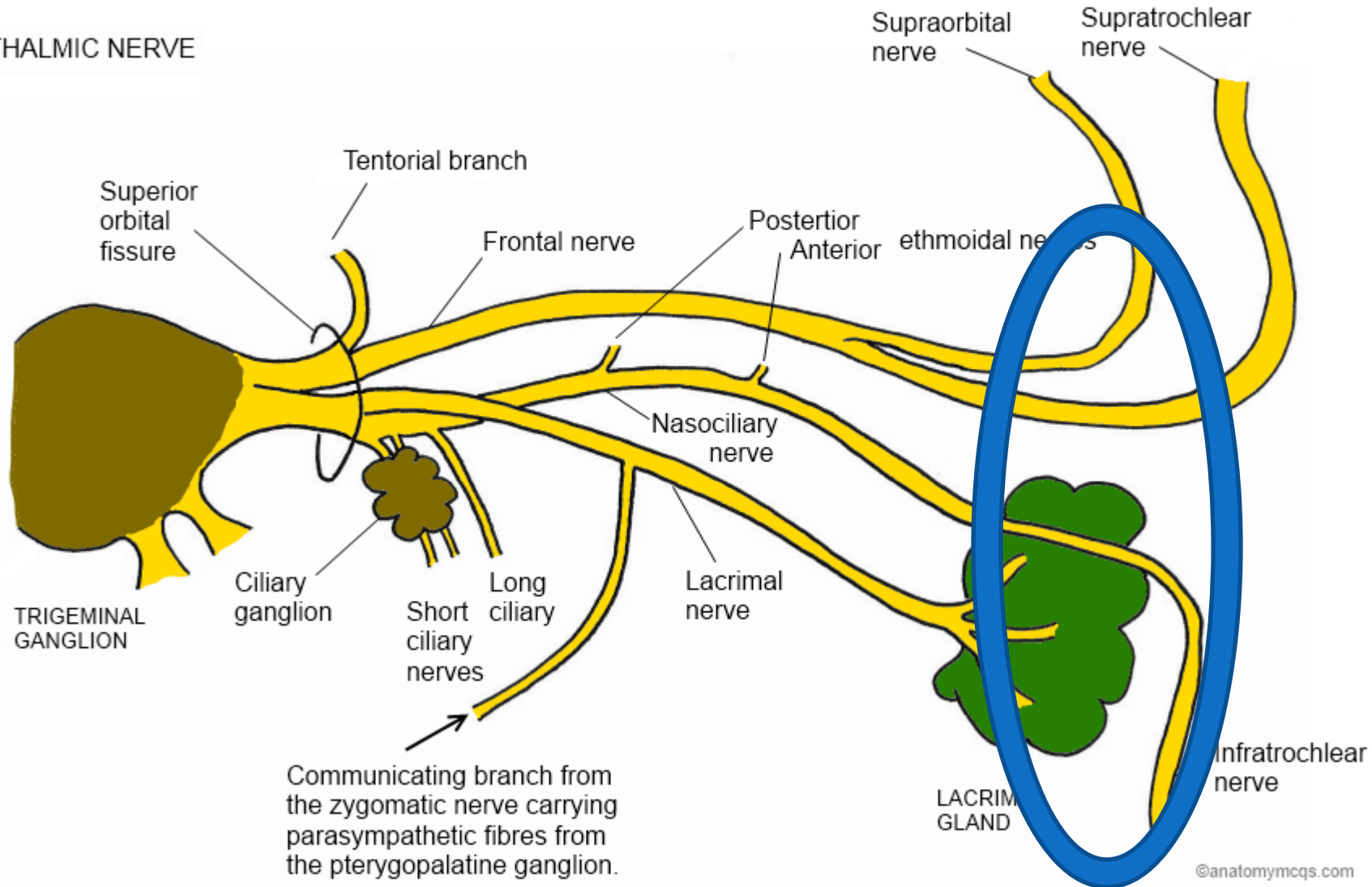
- Smallest division. Only sensory fibres
- Supplies : cornea, conjunctiva, upper lid, forehead, ant part of scalp, nose.
- Course: emerges from trigeminal ganglion at the lateral wall cavernous sinus

3 branches in anterior part of cavernous sinus
lacrimal, nasociliary, frontal

Enters the orbit through superior orbital fissure

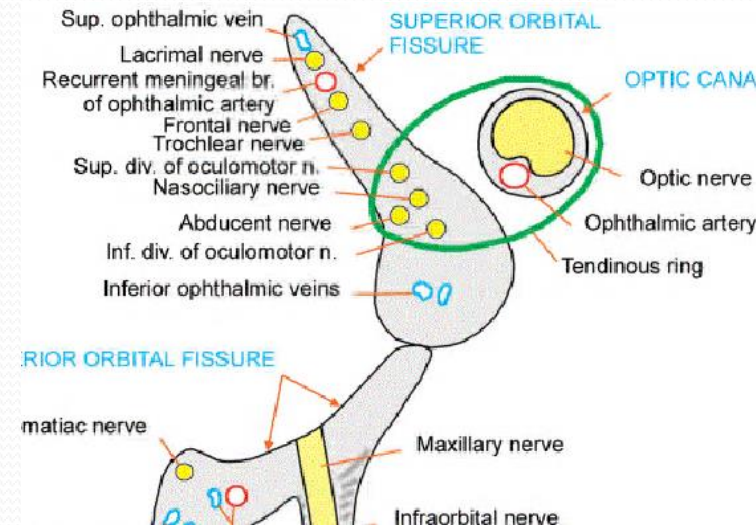


OPHTHALMIC NERVE



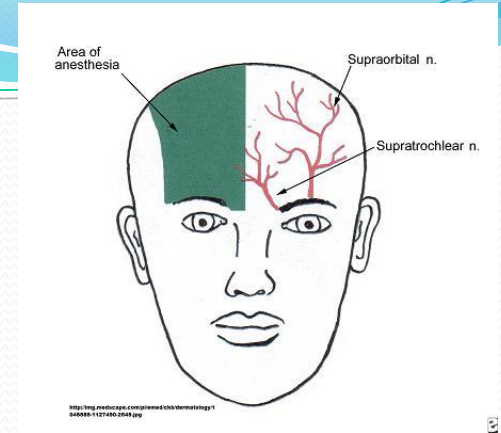
Lacrimal nerve

- smallest branch
- Passes into orbit through lateral compartment of the superior orbital fissure outside the tendinous ring.
- Receives branch from zygomaticotemporal nerve
- Sensory to lateral conjunctiva,
- upper lid, lacrimal gland
- (parasympathetic secretomotor).



Frontal nerve

- Largest
 - Enters through lateral part of superior orbital fissure outside tendinous ring
 - Passes forward between roof of orbit and Levator palpebrae superioris muscle
 - Divides midway into SUPRATROCHLEAR NERVE
 SUPRAORBITAL NERVE
- supplies the conjunctiva, upper eye lid and scalp



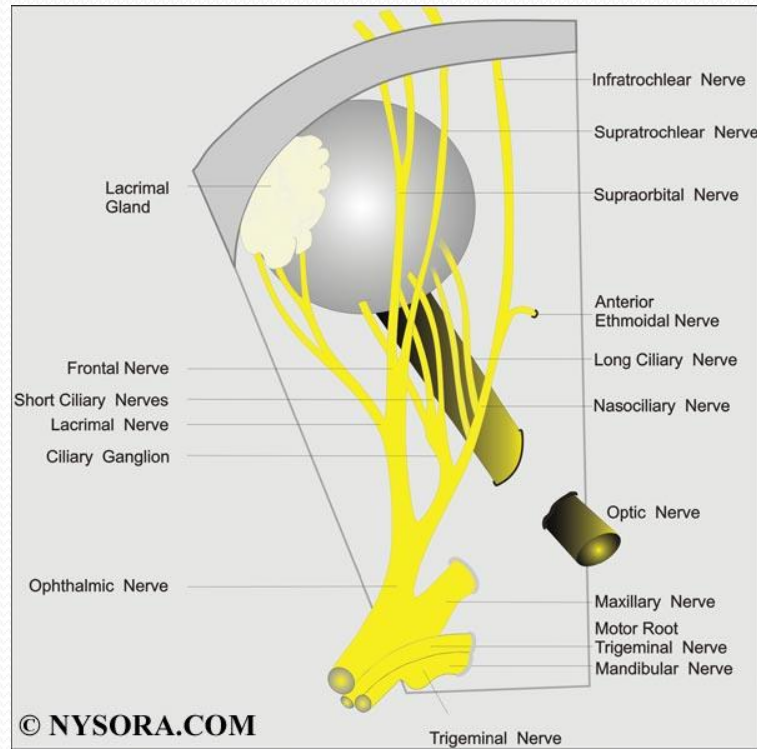
Naso ciliary nerve

- The **nasociliary nerve** travels along the medial border of the orbital roof, giving branches to nasal cavity and ending in the skin at the root of the nose.
- It then branches to **anterior ethmoidal and external nasal nerves**.
- **The ciliary nerves** supplies the iris and cornea
- The **infratrochlear nerve** supplies the skin over the lacrimal sac

The **posterior ethmoidal nerve** supplies the ethmoidal and sphenoid sinuses

The **external nasal nerve** supplies the skin over the apex(tip) and ala of the nose.

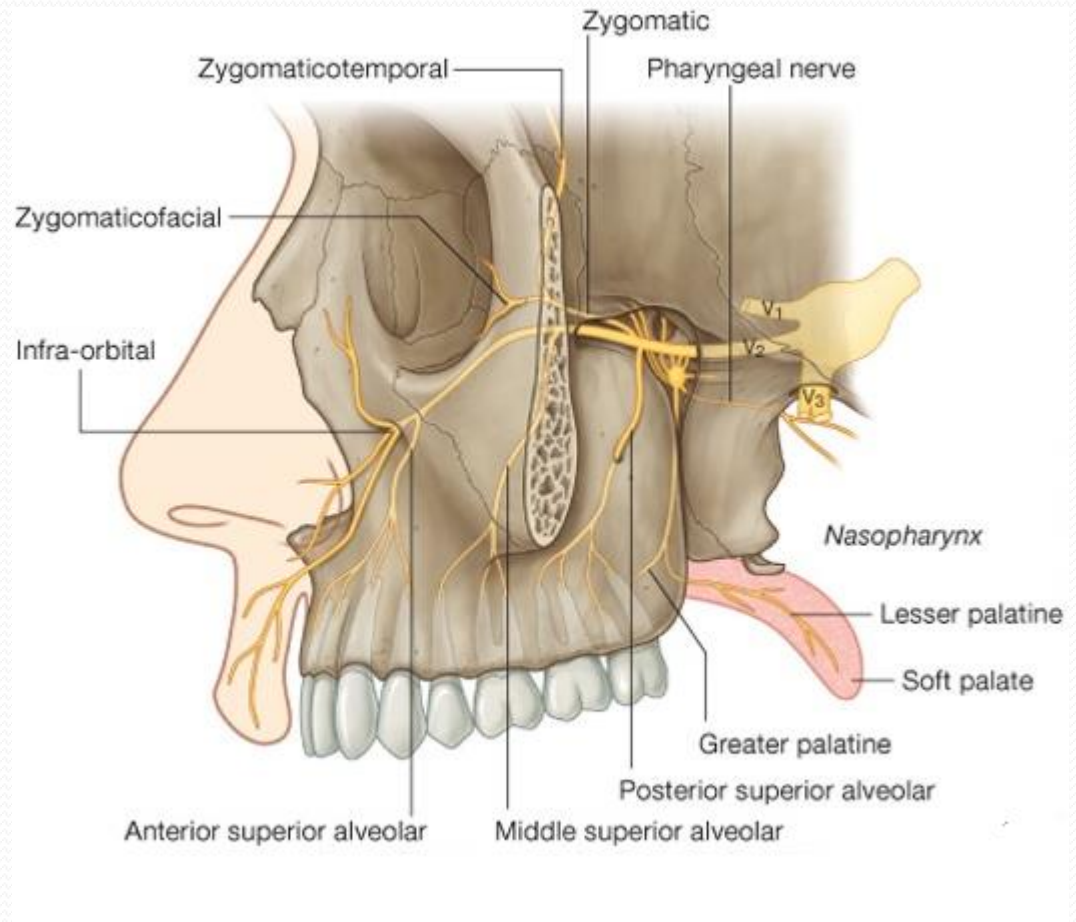
Naso ciliary nerve



MAXILLARY NERVE V2

- **Second division of trigeminal nerve**
- Pure sensory nerve
- The maxillary nerve passes horizontally forward ,leaving the cranium through foramen **rotundum**.
- Outside the cranium ,the maxillary nerve crosses the uppermost part of pterygo palatine fossa.
- As it crosses the pterygo palatine fossa,it gives off branches to **pterygopalatine ganglion, the zygomatic nerves and the posterior superior nerves**.
- It passes through infra orbital groove and becomes **infra orbital nerve** emerges through infraorbital foramen.

Maxillary nerve branches



- **Branches within the cranium:** middle meningeal nerve travels along middle meningeal artery to sensory innervations to the duramater.
- **Branches in pterygopalatine fossa:**
- **1.the zycomatic nerve** branches off at pterygopalatine fossa and enters the orbit through inferior orbital fissure, divides into zycomatico temporal and zycomatico facial nerves-
 - - the zycomatico temporal supplies the sensory innervation to the skin of the forehead.
 - - the zycomaticofacial supply the skin over the prominence of the cheek.

- **2.pterygopalatine nerves(gangloinic branches)**

two short trunks serve as a communication to pteygopalatine ganglion.

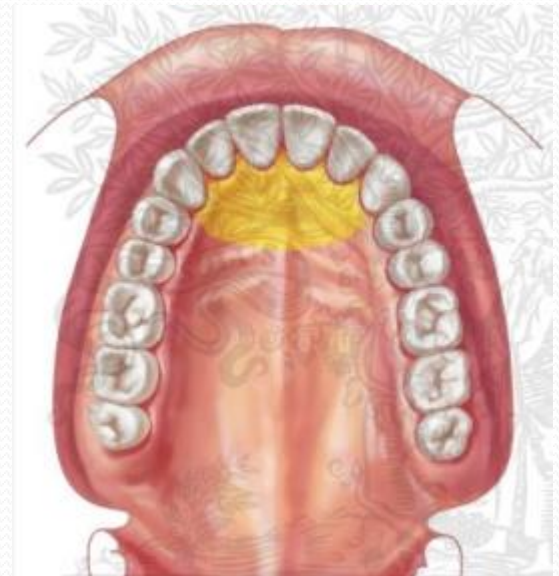
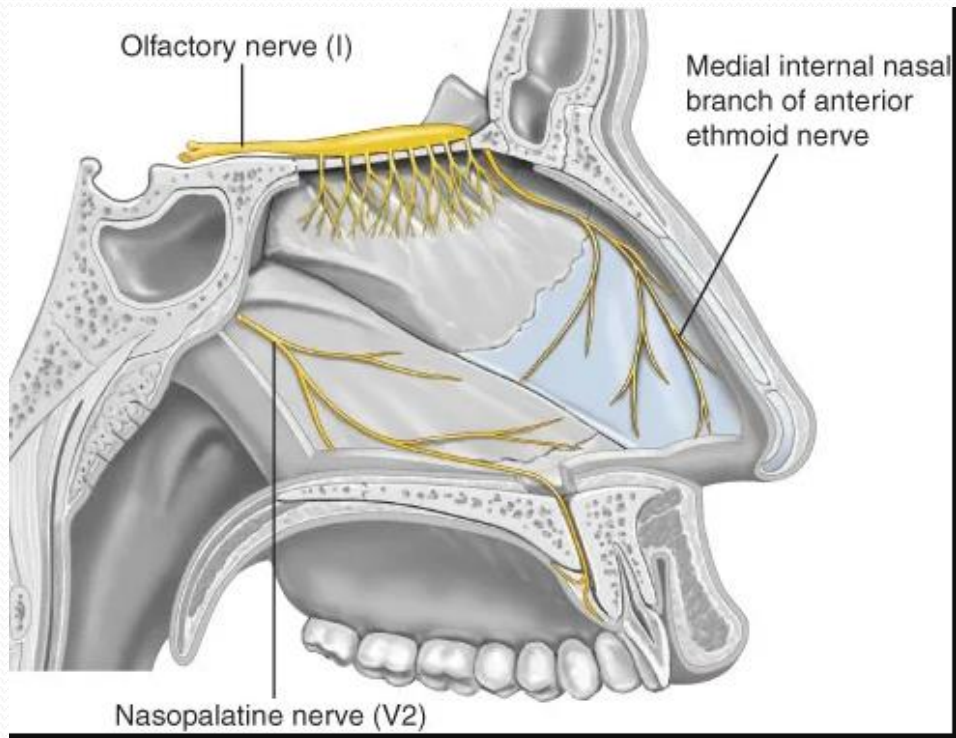
- branches are

a)Orbital branches supply the periosteum of the orbit.

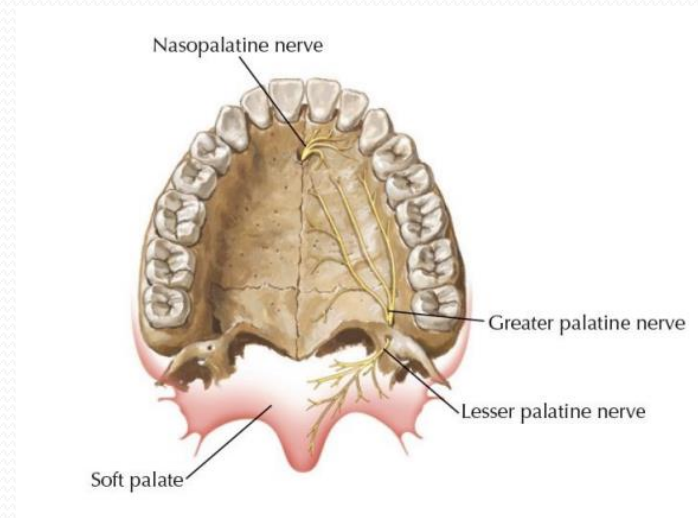
b)Nasopalatine nerve- passes across the roof of the nasal cavity, continoues downward and reaches the floor of the nose gives branches to nasal septum and floor of the nose.

it then enters the incisive canal and emerges in the palate at the incisive foramen, supplies pre maxilla

Nasopalatine nerve



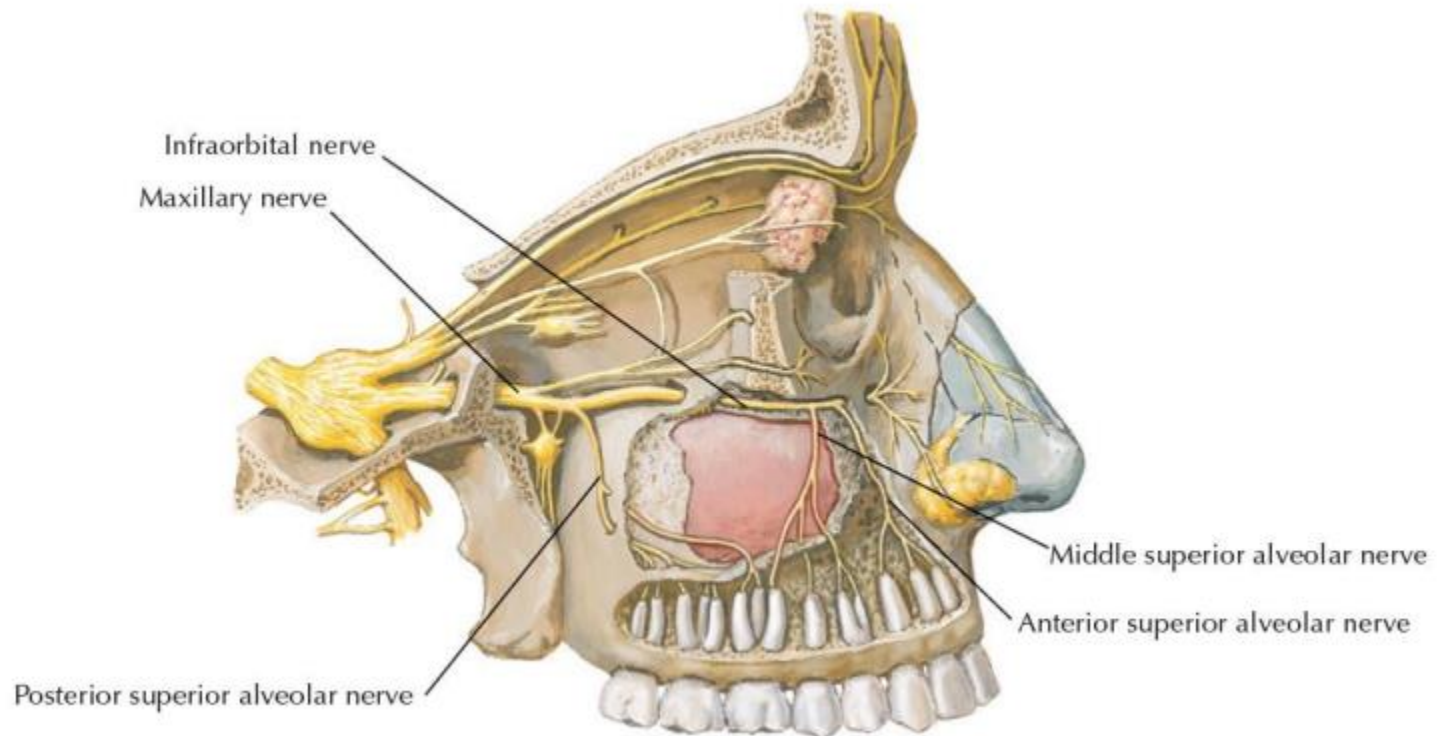
- c) The palatine nerves **greater and lesser palatine nerves** -the **greater palatine nerve** descends through the pterygopalatine canal emerges at the hard palate through greater palatine foramen (distal to second molar) and supplies the posterior palate the **lesser palatine nerve** passes to lesser palatine foramina and supplies the soft palate
- d) **pharyngeal nerve** passes posterior to pharyngeal canal to supply the nasopharynx.



- **3.posterior superior alveolar nerve(PSA)**

- The posterior superior alveolar nerve descends from the main trunk in the pterygopalatine fossa just before the maxillary nerve passes to the infra orbital canal.
- It passes downward the nerve reaches the posterior maxilla
- supplies the maxillary sinus,the molar teeth, bone and periosteum ,gingiva of the molars

Anterior ,Middle & Posterior superior alveolar nerves



Branches in the infraorbital canal

1. middle superior alveolar nerve:

runs along lat wall of maxilla, Participates in superior dental plexus

Supplies premolars teeth and mesiobuccal root of upper first molar .

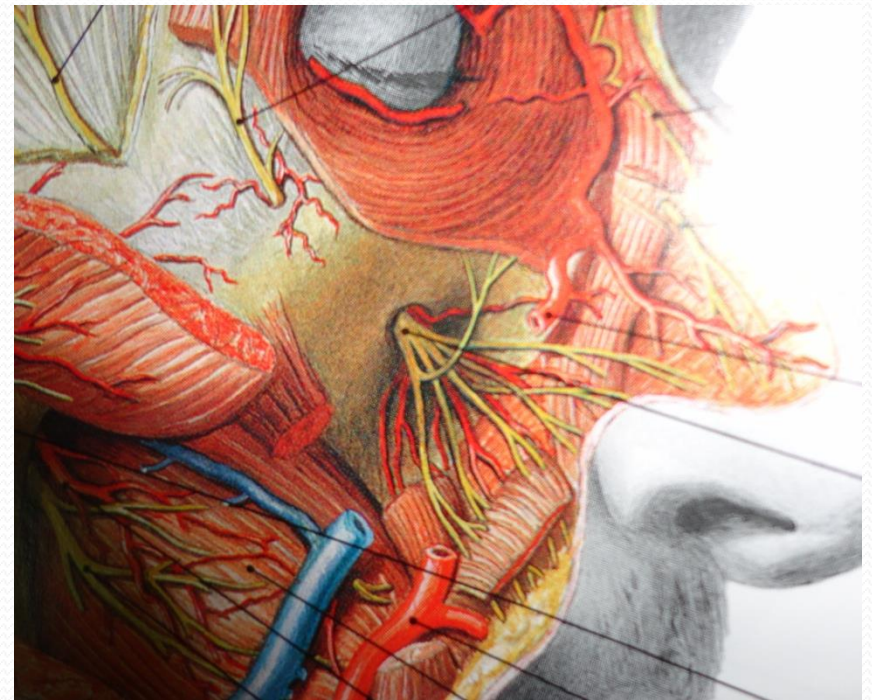
2. Anterior superior alveolar nerve:

Runs in canal in ant wall of maxilla

gives pulpal innervation to incisors and the canine teeth ,bone and the labial mucoperiosteum

Branches in the face

1. **palpebral nerves**-pierces orbicularis oculi and supplies skin of lower lid.
2. **nasal branches**-supplies skin of lat wall nose and mobile part of septum.
3. **superior labial nerve**-supplies skin and mucous membrane of upper lip



MANDIBULAR NERVE V3

- Largest branch
- Mixed- both sensory and motor fibres
- **Motor root**- from motor nucleus in pons
- **sensory root**- gasserian ganglion

both exit through **foramen ovale** in greater Wing of sphenoid

unites to form single trunk **in infratemporal fossa**

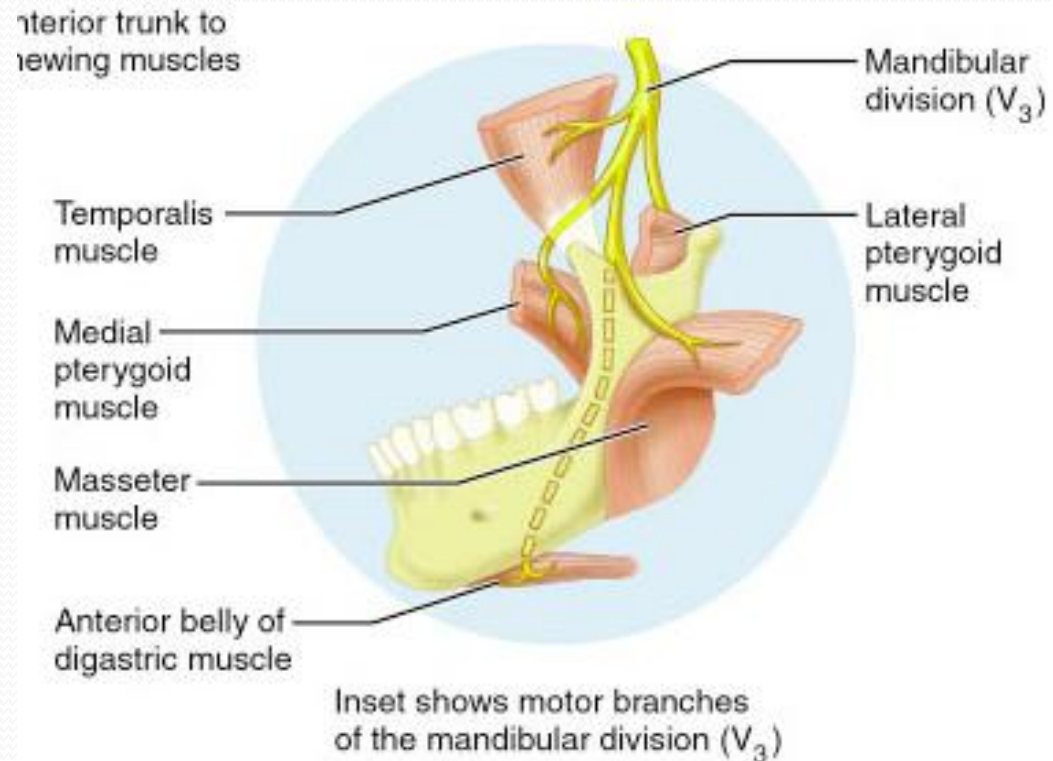
travels between lateral Pterygoid muscle and otic ganglion laterally and tensor palatine muscle medially ,divides

small anterior Division

large posterior division

Motor nerve supply

MOTOR Nerve :
innervates muscles
of mastication and
tensor tympani ,
tensor palatini
mylohyoid and
anterior belly of
digastric.



Branches of V3

Trunk

- Nervous spinosus,
- Nerve to medial Pterygoid

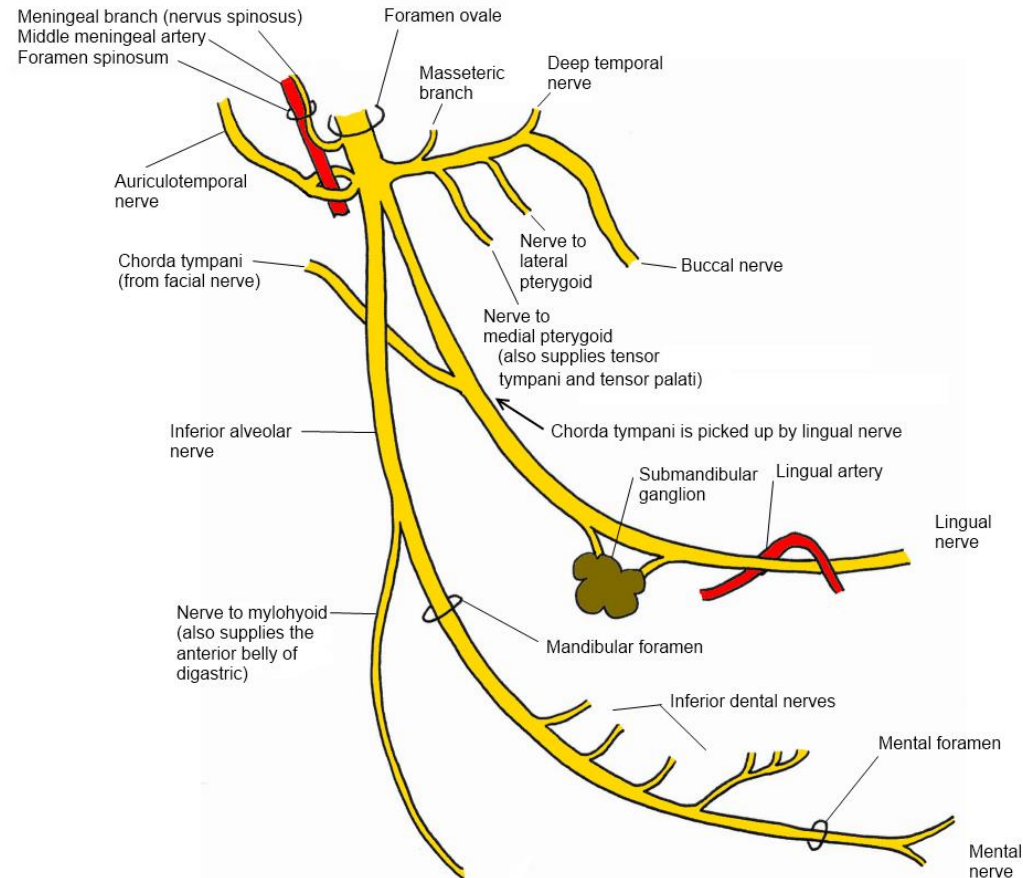
Anterior Division

- Massetric Nerve.,
- Deep temporal Nerve.
- Nerve to lateral Pterygoid
- Buccal Nerve.

Posterior Division

- Auriculo temporal nerve
- Inferior Alveolar nerve
- Lingual Nerve.

MANDIBULAR NERVE



Branches from trunk

- **Nervous spinosus**

- Through foramen spinosum enter the cranial cavity
- supplies the Duramater of mid cranial fossa

- **Nerve to medial Pterygoid**

- Supplies medial pterygoid muscle
- Through otic ganglion without interruption innervation to
 - Tensor tympani muscle
 - Tensor palatini muscle

Branches from the anterior division

- **Nerve to lateral pterygoid**
- **Massetric nerve** supplies masseter and TMJ
- **Deep temporal nerve**-the 2 nerves ascend deep to lateral pterygoid and supply temporalis muscle.
- **Buccal nerve**-is the only sensory branch of anterior division travels between two heads of lateral pterygoid muscle and emerges in` cheek at anterior border of masseter. Supplies skin and mucous of cheek and also buccal gingiva upto mesial of 1st mandibular molar tooth

Branches from the posterior division

1. Auriculotemporal nerve-

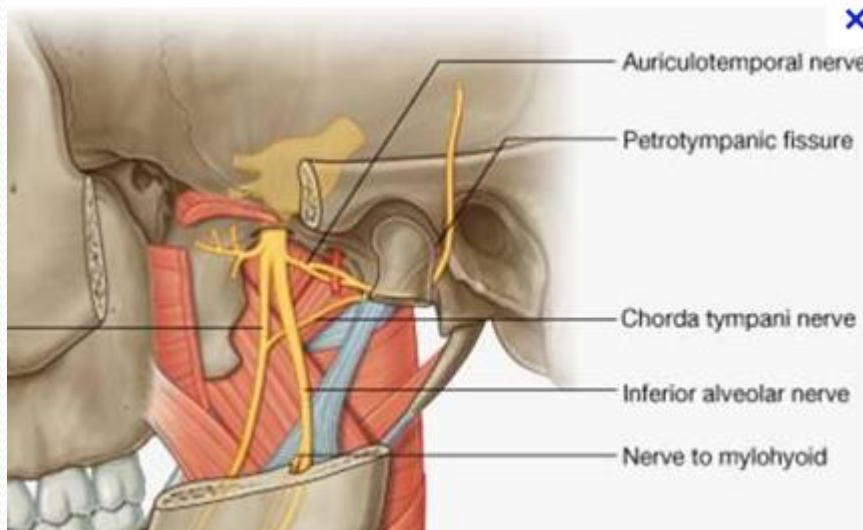
Arises from 2 roots which encircle the middle meningeal artery

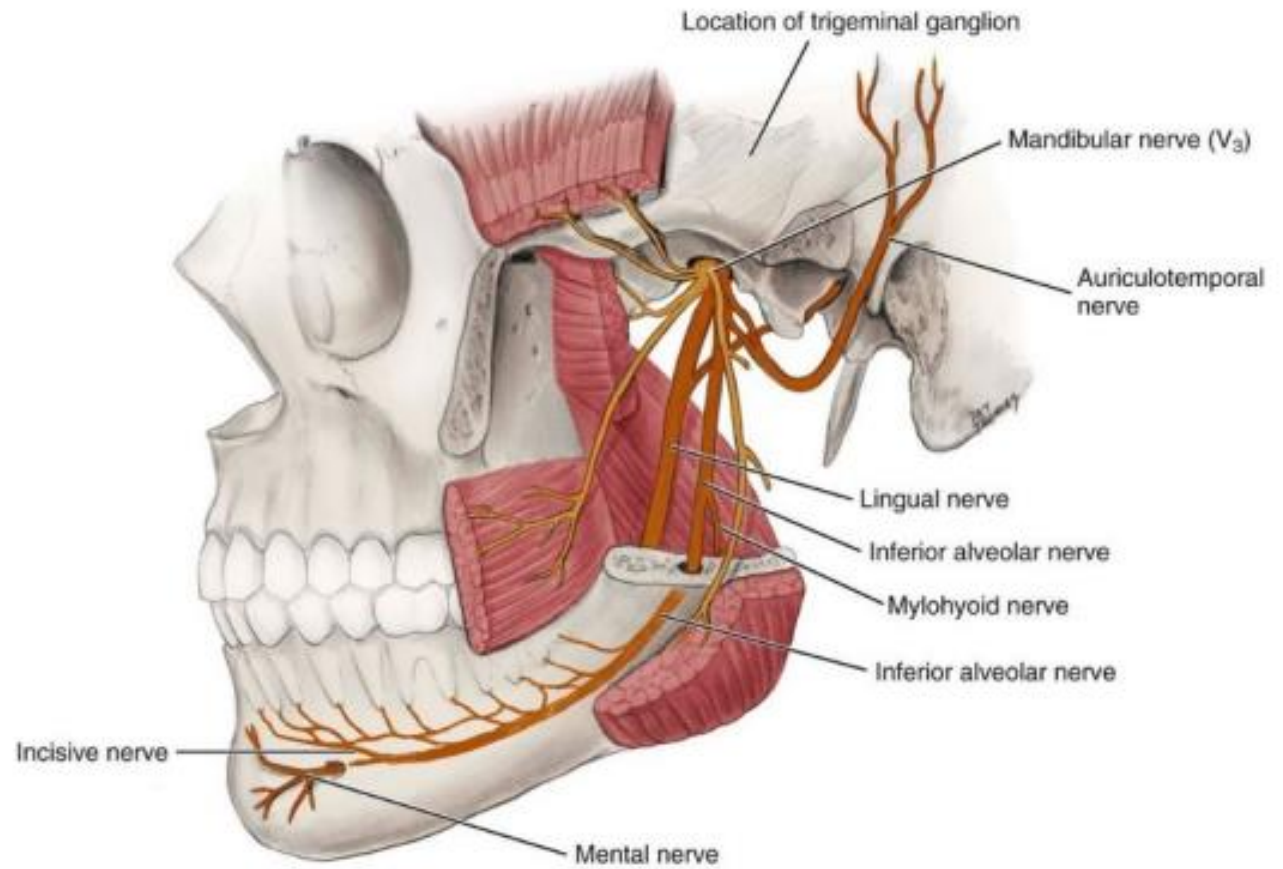


Lies behind the TMJ close to the parotid



Ascends behind superficial temporal vessels and then in temporal region divides into superficial temporal branches.





2. Inferior alveolar nerve:

Is mixed nerve

Passes between mandible and sphenomandibular ligament
lateral to medial pterygoid,

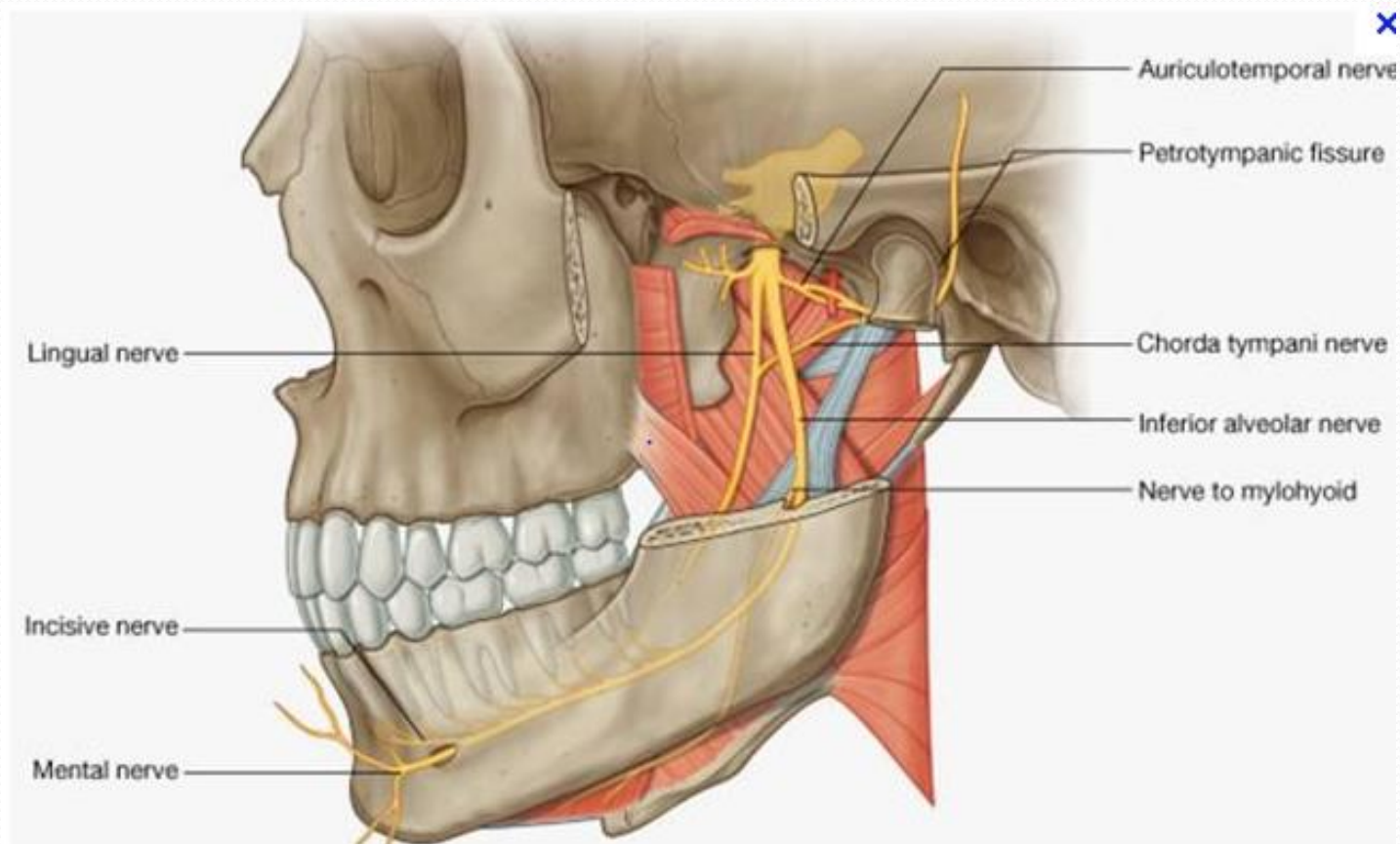
Enters mandible through mandibular foramen to run in a bony canal below the teeth

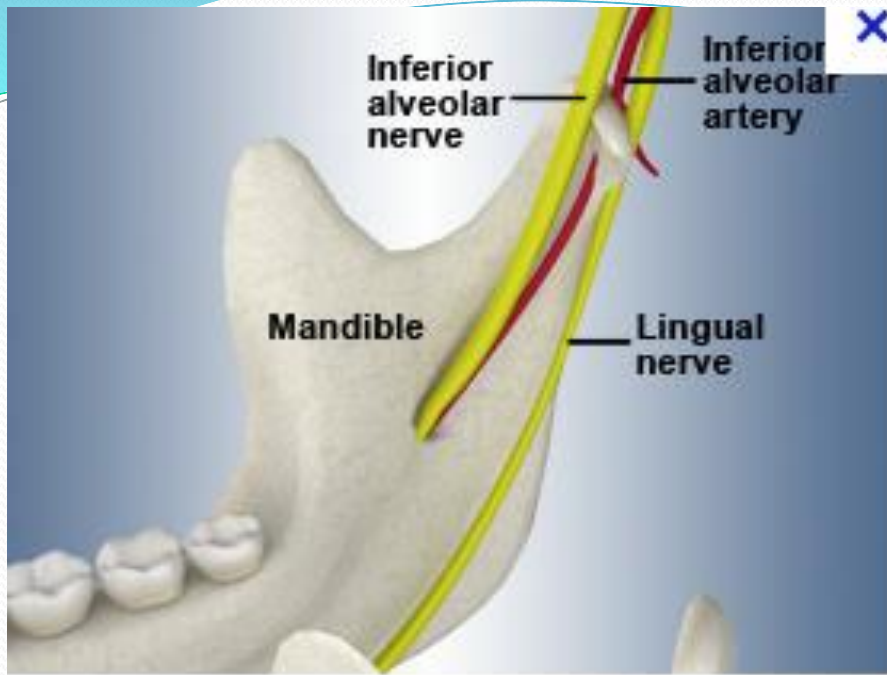
Branches: to molars and premolars pulpal innervation,
incisive nerve to incisors and canine tooth

mental nerve comes out through mental foramen, supply lower lip, labial gingiva

gives a branch mylohyoid nerve before it enters mandibular foramen- innervates **mylohyoid and anterior belly of digastric muscles**

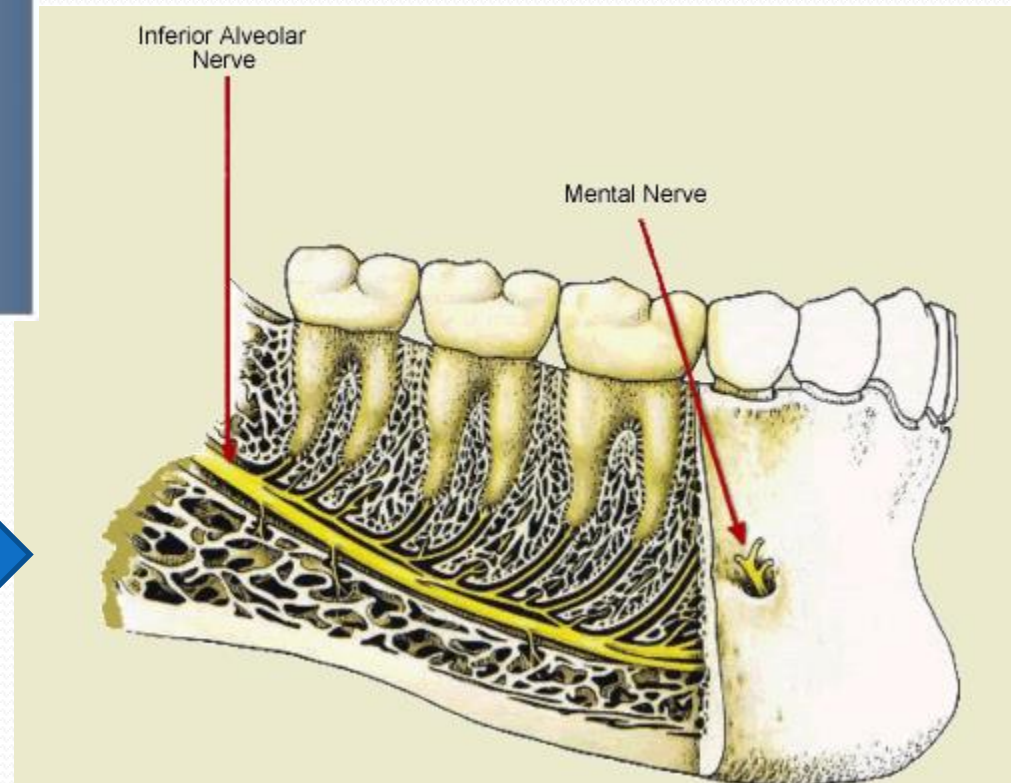
Inferior alveolar nerve





Inferior alveolar nerve enters mandibular foramen

Inferior alveolar nerve runs below the molar, premolar teeth in mandibular canal



Inferior alveolar canal in OPG x ray



Damage of inferior alveolar nerve during surgical removal of wisdom tooth



3.Lingual nerve:

lies anterior to inferior alveolar nerve between lateral pterygoid and tensor palatini muscles

receives chorda tympani from facial nerve(taste)

Emerges from inferior border of lateral pterygoid to lie between ramus and medial pterygoid

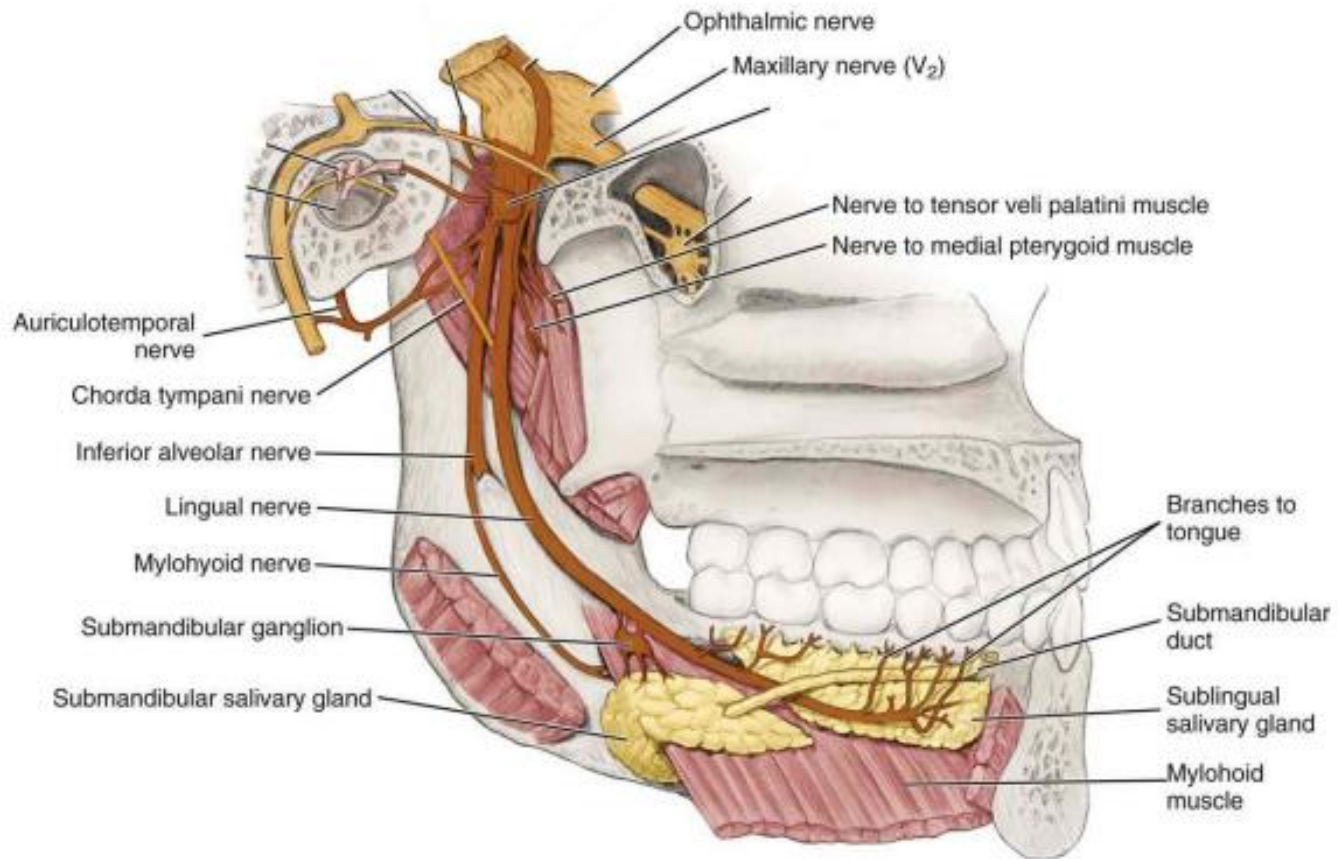
passes downwards close to mandible

1 cm below and behind 3rd molar in gingiva

passes above hypoglossus lateral to the tongue where it is related to the submandibular ganglion

Gives sensory supply to tongue ,floor of mouth, mandibular gums and carries proprioception from tongue.

Lingual nerve

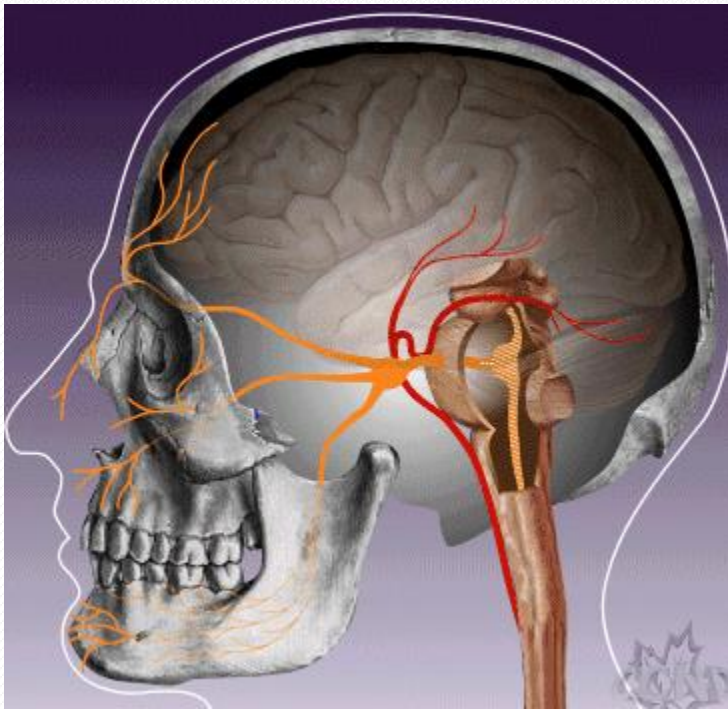


Applied anatomy:

Trigeminal neuralgia

- Trigeminal neuralgia is a neurological disorder characterized by episodes of intense **pain in the face, originating from the trigeminal nerve**
- patients may describe a **trigger area** on the face so sensitive that touching or even air can trigger an episode; electric shock like pain , lasts only seconds
- Seen in older patients , mostly in females

Compression of trigeminal nerve root by a branch of superior cerebellar artery



- research indicates that it is an enlarged blood vessel - possibly the **superior cerebellar artery** - compressing against the the trigeminal nerve near its connection with the pons
- Such a compression can injure the nerve's protective myelin sheath and cause hyperactive functioning of the nerve. This can lead to pain attacks at the slightest stimulation



Completion of Trigeminal nerve

THANK YOU